

Self-Learning: Topological Sorting Data Structures

> **Definition:** Topological Sorting of a **Directed Acyclic Graph (DAG)** is a linear ordering of vertices such that for every directed edge (u, v) , vertex u comes strictly **before** vertex v in the ordering.

1. Detailed Technical Explanation

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Valid Topological Orders:

- 5 ->

Key Conditions:

1. Topological sorting is defined **ONLY** for DAGs (Directed Acyclic Graphs).

2. If a

2. Algorithms for Topological Sorting

1. Kahn's Algorithm (BFS In-Degree Approach)

1. Compute the **In-Degree** of every vertex in the graph.

2. Pus

2. DFS with Stack Approach

1. Initialize a `visited[]` boolean array.

2. For

3. Real-World Applications

- **Build Systems (Make, Gradle, Webpack):** Resolving source code compilation dependencies.

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4. Quick Recall Flow

DAG C